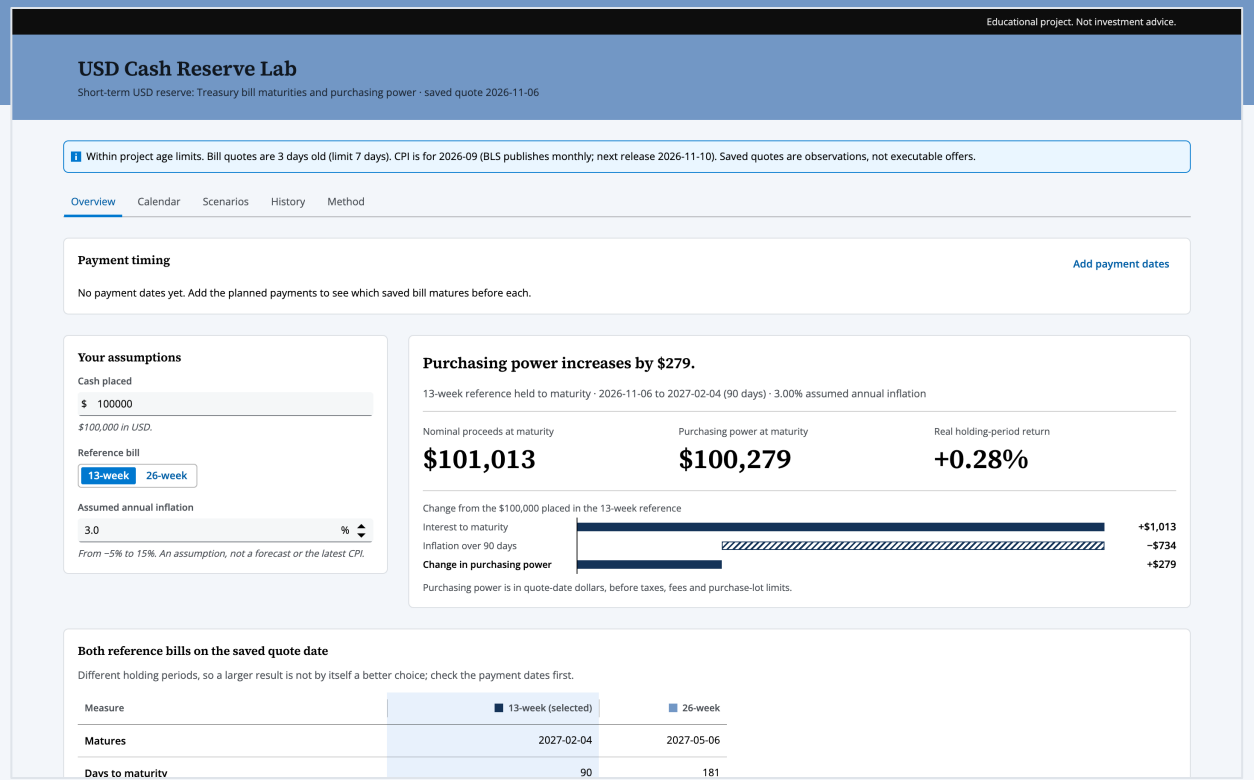
Backup slides

USD Cash Reserve Lab

When is the cash there for each payment,
and what does inflation do to it meanwhile?



Group 9 · BMS5312 Fintech Management
Live dashboard demo, 5 minutes

Speaker 1 · The user and the question

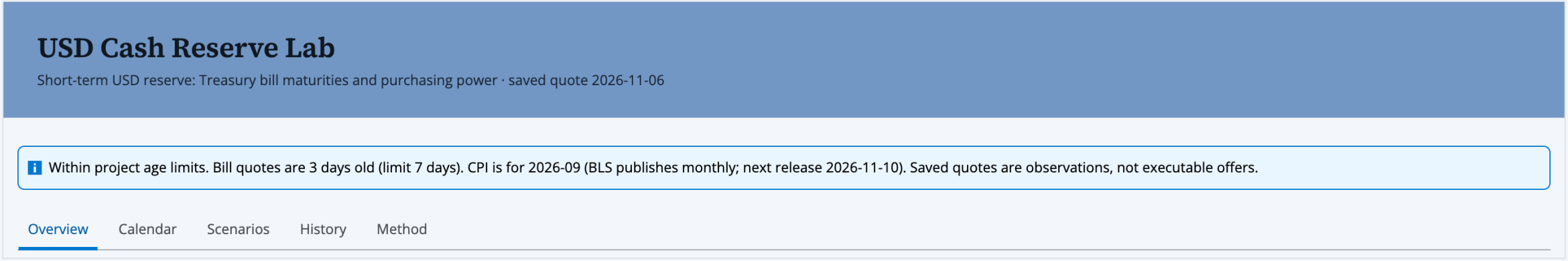
Who is the user, and what is the page for?

The user

- A family-office client and their advisor
- A US-dollar reserve for payments in the next few months: tax, tuition, a household budget
- **Today:** spreadsheets and separate data websites

A benchmark for the reserve

- **Timing:** will the cash be there on time? The calendar checks all seven saved bill maturities.
- **Value:** what happens to purchasing power in the safest short-term bills? Two reference bills, 3-month and 6-month, the maturities the Federal Reserve publishes in H.15.
- Which bill to buy, and anything beyond the reserve, stays with the advisor.



Educational project. Not investment advice.

Speaker 2 · Live dashboard

When is the cash there for each payment?

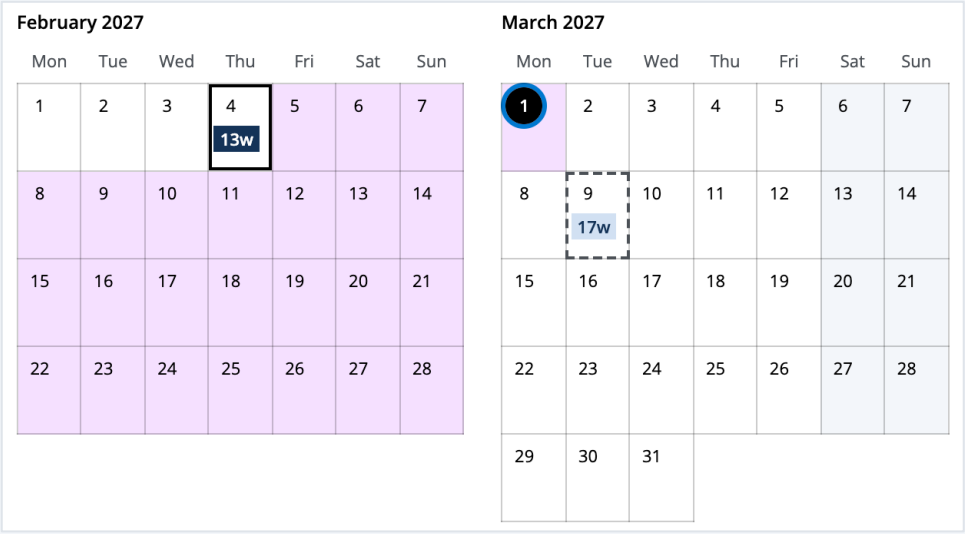
Tuition, Monday 1 March 2027

Fixed USD payment

13w **Matures on or before:** 13-week bill, 2027-02-04

Cash waits 25 days, no return assumed

17w **Next maturity:** 17-week bill, 2027-03-09, 8 days late
So it would have to be sold before maturity at an unknown price



Calendar fit by payment

Payment	Date	Matures on or before	Cash waits	Next maturity after
Property tax instalment	2027-01-15	8-week · 2027-01-05	10 days	13-week · 2027-02-04, 20 days after
Tuition	2027-03-01	13-week · 2027-02-04	25 days	17-week · 2027-03-09, 8 days after
Spring household budget	2027-04-15	17-week · 2027-03-09	37 days	26-week · 2027-05-06, 21 days after

Speaker 2 · Live dashboard

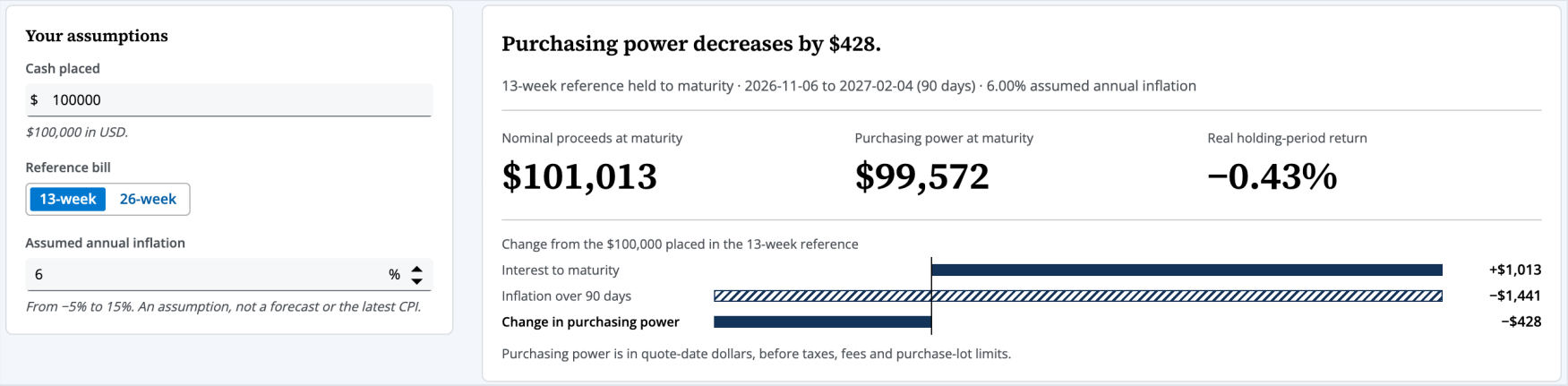
What does inflation do to the cash?

$100,000 in the 13-week bill · 2026-11-06 to 2027-02-04 (90 days) · bank discount quote 4.01%

	Inflation 3%	Inflation 6%
Nominal proceeds at maturity	$101,013	$101,013
Purchasing power at maturity	$100,279	$99,572
Real holding-period return	+0.28%	−0.43%
Change in purchasing power	**+$279**	**−$428**

> Inflation changes what the money buys, not the dollars that arrive.

Break-even inflation:
13-week 4.17%, 26-week 4.31%.
A threshold, not a forecast.



Live page, inflation set to 6%

Speaker 3 · Data and method

How do we know the numbers are right?

Price per $100

$$P = 100\,(1 - d \cdot n/360)$$

Nominal proceeds

$$N = B \cdot 100/P$$

Inflation factor

$$F = (1 + a)^{n/365}$$

Purchasing power

$$R = N/F$$

Independent Excel check

By a member who did not build the page · quote 2026-11-06 · inputs copied from the official websites

Check	Excel	Page	Result
Price per $100	98.9975	—	*input to the next rows*
Nominal proceeds	101,012.65	$101,013	**PASS**
Purchasing power at 3%	100,279.10	$100,279	**PASS**
Real holding-period return	0.279%	+0.28%	**PASS**
Break-even, 13-week	4.1708%	4.17%	**PASS**
Break-even, 26-week	4.3114%	4.31%	**PASS**
CPI year on year, Sep 2026	3.7377%	3.74%	**PASS**

6 / 6
checks passed

Limitations

- Saved quotes are observations, not executable prices.
- Settlement, taxes and fees are not modelled.
- Inflation is an assumption, not a forecast.

Speaker 4 · AI maintenance

How does an AI assistant update the data, and who decides?

AI assistant

1. Fresh AI chat reads SKILL.md
2. Downloads into candidate/
3. Rebuilds and pre-reviews
4. Before/after table with source links

Person

5. Person compares and accepts
6. Release; old version kept; checks rerun

Normal update, 2026-11-09

	Before	After
Treasury quote date	2026-10-26	**2026-11-06**
13-week bank discount	4.05%	**4.01%**
26-week bank discount	4.21%	**4.12%**
CPI month	2026-09	**2026-09** (no new release)

Failure test

Rates set to basis points
→ **STOP: Incompatible input unit**
Nothing written; data.js unchanged.

The AI does the checks; the source comparison and the decision stay with a person.

Speaker 5 · Handover

Can someone else run it after the course?

Handover

- The ZIP alone works
- Double-click index.html
- One command rebuilds data.js byte for byte
- One command downloads and checks an update
- Nothing is released until a person accepts
- The previous version is one command away

Risks

- No scheduler, so quotes go stale after 7 days and the page says “Review required”
- BLS release dates must be added each December
- October 2025 CPI was never published

TO UPDATE AFTER THE USER TEST

Feedback and the change we made

AI-assisted review by a teammate: the timing chart shows seven maturities, but the value uses two reference bills. The chart caption now names the two reference bills valued below.

Next improvement

Let the value follow whichever bill fits each payment, not only the two reference bills.